THE STACK OF FINITE TYPE SPACES IS FIBRED IN GROUPOIDS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Section examples-stacks-section-stack-in-groupoids-of-finite-type-spaces (canonical tag 04UH). cited at proof lines 3494-3496. Prior treatment.
- Definition categories-definition-fibred-groupoids (canonical tag 003T). cited at proof line 3500. The conditions.
- Lemma spaces-morphisms-lemma-base-change-finite-type (canonical tag 03XH). cited at proof line 3507. Finite type of the pullback.
- Lemma spaces-morphisms-lemma-base-change-finite-presentation (canonical tag 03XR). cited at proof line 3512. fp case.
- Lemma spaces-morphisms-lemma-base-change-flat (canonical tag 03MO). cited at proof line 3513. Flat case.
- Lemma spaces-morphisms-lemma-base-change-proper (canonical tag 04WP). cited at proof line 3514. Proper case.

Source reference. Original section 04UH: sources/stacks-chow/tags.txt, source lines 6290–6290. Cited at proof lines 3494-3496: prior treatment.

Source reference. Original definition 003T: sources/stacks-chow/tags.txt, source lines 147–147. Cited at proof line 3500: the two conditions.

Source reference. Original lemma 03XH: sources/stacks-chow/tags.txt, source lines 5148–5148. Cited at proof line 3507: X' -> S' finite type.

Source reference. Original lemma 03XR: sources/stacks-chow/tags.txt, source lines 5158–5158. Cited at proof line 3512: fp case.

Source reference. Original lemma 03MO: sources/stacks-chow/tags.txt, source lines 4756–4756. Cited at proof line 3513: flat case.

Source reference. Original lemma 04WP: sources/stacks-chow/tags.txt, source lines 6367–6367. Cited at proof line 3514: proper case.

Lemma 0.2. *The category $\mathcal{S}paces'_{ft}$ is fibred in groupoids over Sch_{fppf}. The same is true for $\mathcal{S}paces'_{fp,flat,proper}$.*

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Proof. We have seen this in Examples of Stacks, Section 04UH for the case of $\mathcal{S}paces'_{ft}$ and this easily implies the result for the other case. However, let us also prove this directly by checking conditions (1) and (2) of Categories, Definition 003T.

Condition (1). Let $X \to S$ be an object of $\mathcal{S}paces'_{ft}$ and let $S' \to S$ be a morphism of schemes. Then we set $X' = S' \times_S X$. Note that $X' \to S'$ is of finite type by Morphisms of Spaces, Lemma 03XH. to obtain a morphism $(X' \to S') \to (X \to S)$ lying over $S' \to S$. Argue similarly for the other case using Morphisms of Spaces, Lemmas 03XR, 03MO, and 04WP.

Condition (2). Consider morphisms $(f, g) : (X' \to S') \to (X \to S)$ and $(a, b) : (Y \to T) \to (X \to S)$ of $\mathcal{S}paces'_{ft}$. Given a morphism $h : T \to S'$ with $g \circ h = b$ we have to show there is a unique morphism $(k, h) : (Y \to T) \to (X' \to S')$ of $\mathcal{S}paces'_{ft}$ such that $(f, g) \circ (k, h) = (a, b)$. This is clear from the fact that $X' = S' \times_S X$. The same therefore works for any full subcategory of $\mathcal{S}paces'_{ft}$ satisfying (1). $\square$